

Jiayi Chen

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EDUCATION

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)

Sept. 2024 – Present

Ph.D. in Computer and Information Engineering, School of Science and Engineering (SSE)

Shenzhen, China

- **Research Area:** Embodied Intelligence
- **Advisor:** Prof. Guangxu Zhu; **Co-advisor:** Prof. Shuai Wang

Beijing University of Posts and Telecommunications (BUPT)

Sept. 2020 – Jun. 2024

B.Eng. in Internet of Things Engineering, Intl. School; QMUL-BUPT Joint Programme

Beijing, China

- **Weighted Average Mark:** 90.07/100; **GPA:** 3.73/4.0; ranked **6/189**
- Selected into the YePeida Institute Pilot Class, admitting approximately the top **7%** of students

PUBLICATIONS

Jiayi Chen, Shuai Wang, Guangxu Zhu, and Chengzhong Xu. “Bridging Large-Model Reasoning and Real-Time Control via Agentic Fast-Slow Planning.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026, accepted. arXiv:2604.01681.

Jiayi Chen, Shuai Wang, Guoliang Li, Wei Xu, Guangxu Zhu, Derrick Wing Kwan Ng, and Chengzhong Xu. “Opportunistic Collaborative Planning with Large Vision Model Guided Control and Joint Query-Service Optimization.” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 951–958.

Jiayi Chen, Shuai Wang, Guangxu Zhu, and Chengzhong Xu. “Not All Uncertainty Matters: Simulation-in-the-Loop Fast–Slow Reasoning for Decision-Critical Autonomous Driving System.” *IEEE Transactions on Mobile Computing*, submitted.

Jiayi Chen, Weiting Ou, and Guangxu Zhu. “The Universal Language of CSI: Unifying Wireless Sensing Across Devices and Environments.” *IEEE Communications Magazine*, accepted.

RESEARCH EXPERIENCE

Agentic Fast-Slow Planning for Real-Time Autonomous Driving

Jan. 2025 – Jan. 2026

First-author work; accepted by ICRA 2026; related patent granted

CUHK-Shenzhen

- Identified the gap between slow reasoning and real-time control; proposed **AFSP** to decouple reasoning, planning, and MPC while mapping VLM/LLM understanding to semantic A* trajectories
- Validated in CARLA, reducing completion time by over **12%** and maximum lateral deviation by up to **45%** against MPC and A*-MPC

Opportunistic Collaborative Planning for Open-World Autonomous Driving

Sept. 2024 – Oct. 2025

First-author work; published at IROS 2025, extended to TMC submission

CUHK-Shenzhen

- Showed cloud large models should be invoked by decision value; proposed **OCP** with LVM-MPC, timing optimization, and simulation-in-the-loop opportunity estimation
- Implemented in CARLA/ROS, reducing service calls by **50%**, improving success rate by over **6%**, and cutting dynamic-scenario time by **26.2%**

Universal CSI Foundation Model for Heterogeneous Wireless Sensing

Sept. 2024 – Present

First-author work; accepted by IEEE Communications Magazine

CUHK-Shenzhen

- Framed CSI heterogeneity as “language” alignment; proposed **WiLLM** and built a **259,774**-sample corpus from **12** datasets and **6** task categories
- Used dataset-specific tokenizers and a shared GPT-style Transformer, reaching **91.06%** accuracy and robust **1%**-label few-shot adaptation

INTERNSHIP EXPERIENCE

AgiBot Robotics

May 2026 – Present

Embodied Intelligence Research Intern; Vision-and-Language Navigation (VLN)

Shenzhen, China

- Developing **MapFusion**, a query-conditioned VLN stack that routes language queries across heterogeneous maps and applies constraint-aware reasoning for grounded, rejectable, reachable goals
- Validated on an AgiBot **KuoTuo D2 Max** through cross-floor real-robot trials, integrating perception, map evidence, and executable goal generation
- Extended toward inspection-oriented embodied applications, including utility/grid-inspection scenarios; preparing an **ICRA** submission

Shanghai AI Laboratory

Jan. 2023 – Dec. 2024

Remote Research Intern; supervised by Xinyu Cai

Remote

- Developed an LLM-based traffic scenario generation pipeline that converts natural-language requirements into **OpenSCENARIO**-compatible simulation cases, with generator-judger agents for initial-point selection, control-code generation, and XML output
- Configured CARLA-Apollo communication and built Waymo-to-V2X-ViT conversion scripts for autonomous-driving simulation; generated high-quality **InternVL** VLM-OCR data through difficult-sample synthesis, rule-based checks, and parameter-freezing fine-tuning

Apple Inc.

Jul. 2022 – Mar. 2023

Data Analyst Intern, Channel Planning and Forecast

Beijing, China

- Built a modular Python/C++ allocation pipeline for weekly iPad distribution, processing **1M+** units for authorized stores in China in about **2 minutes**
- Reduced allocation errors from roughly **40%** to full correctness and improved AOS product tracking speed from **20s** to **5s** per product
- Automated pricing, reporting, and launch workflows with web crawlers, Excel-driven templates, scheduled scripts, email delivery, and Tableau dashboards

PROJECT EXPERIENCE

6G Network Digital Twin and Ray-Tracing Wireless Simulation Platform

Sept. 2024 – Present

Project participant, Guangdong Major Project on Environment-Enhanced 6G Networks

CUHK-Shenzhen

- Built and diagnosed a **Sionna RT**/Taichi RT pipeline for 6G digital twins, covering OSM/GPS alignment, ray/path extraction, benchmarks, and visualization
- Designed semantic-aware calibration with LiDAR maps, SAM segmentation, sparse RSRP fitting, antenna sampling, and phase-aware pruning

Multi-Sensor UAV Docking and Adaptive Cruise Control System

Oct. 2021 – Jan. 2023

Project Leader; supervised by Prof. Haiyuan Li

- Led the design of a multi-sensor fusion system for UAV aerial docking and adaptive cruise control
- Implemented PnP-based visual recognition for mid-range positioning, enabling centimeter-level docking accuracy
- Coordinated integration, validation, and competition materials; won university and Internet+ municipal third prizes

AWARDS AND HONORS

Chinese Invention Patent (AFSP/ICRA-related): “Method and Related Device for Autonomous-Driving Path Planning Integrating Large-Model Reasoning,” ZL 2026 1 0345449.4.

Outstanding Undergraduate Thesis, Queen Mary-BUPT Joint Programme: WiFi CSI Action Recognition

Jun. 2024

The 8th China International “Internet+” College Student Innovation and Entrepreneurship Competition, BUPT Industrial Track Second Prize

Jul. 2022

First Prize, National College Mathematics Competition

Jul. 2021

Programming Contest Organized by SenseTime, Bronze Prize (Top 10 Team)

Dec. 2020

The 15th Programming Design Contest of BUPT, Bronze Prize (Top 10 Team)

Mar. 2021

ENGLISH PROFICIENCY

English: TOEFL iBT 103/120 (R27/L28/S23/W25), CET-6 516, CET-4 607

ACADEMIC SERVICE

Reviewer: ICRA 2025/2026, IROS 2025, ICML 2026, SIGIR 2026, NIPS 2026, ICC 2025; IEEE TMC, IEEE WCL